SkillGauge AI – Interview Assessment Report

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1. In Python, which built-in function is primarily used to iterate over two or more iterables simultaneously?
A) map()
B) zip()
C) reduce()
D) filter()
Student Answer: C
Correct Answer: B
Marks: 0/1

2. Which of the following data structures in C guarantees that elements are stored contiguously in memory?
A) Linked List
B) Array
C) Binary Search Tree
D) Queue implemented using pointers
Student Answer: C
Correct Answer: B
Marks: 0/1

3. In SQL, which clause is used to filter results based on aggregate functions (like COUNT, SUM, AVG)?
A) WHERE
B) GROUP BY
C) ORDER BY
D) HAVING
Student Answer: C
Correct Answer: D
Marks: 0/1

4. Which HTML5 semantic element is best suited for wrapping the main content area of a document, excluding headers, footers, and sidebars?
A) <section>
B) <article>
C) <main>
D)
Student Answer: C
Correct Answer: C
Marks: 1/1

5. In CSS, what is the correct order of precedence (highest to lowest) for applying styles, assuming they are defined in different places?
A) Inline Style > ID Selector > Class Selector > Element Selector
B) ID Selector > Inline Style > Class Selector > Element Selector
C) Element Selector > Class Selector > ID Selector > Inline Style
D) Class Selector > ID Selector > Element Selector > Inline Style
Student Answer: A
Correct Answer: A
Marks: 1/1

6. In a standard Machine Learning Pipeline, which step immediately follows Feature Engineering?
A) Model Deployment
B) Data Ingestion
C) Model Training
D) Data Visualization
Student Answer: A

Correct Answer: C
Marks: 0/1

7. When evaluating a binary classification model on an imbalanced dataset, which metric is generally considered more reliable than simple Accuracy?
A) Precision
B) Recall
C) F1 Score
D) Support
Student Answer: A
Correct Answer: C
Marks: 0/1

8. What is the primary function of ‘git stash’ in Github/Git workflow?
A) To discard all local uncommitted changes permanently.
B) To temporarily save uncommitted changes without creating a commit.
C) To synchronize local changes with the remote repository.
D) To merge changes from one branch into another.
Student Answer: C
Correct Answer: B
Marks: 0/1

9. Which VS Code feature allows developers to work seamlessly on code hosted on remote SSH servers or within containers?
A) IntelliSense
B) Remote Development Extensions
C) Live Share
D) Task Runner
Student Answer: C
Correct Answer: B
Marks: 0/1

10. Dataiku DSS (Data Science Studio) is primarily known for what capability?
A) High-performance data warehousing (storage)
B) Low-code/No-code platform for managing the entire data science lifecycle.
C) Interactive 3D geospatial visualization.
D) Managing Docker container orchestration.
Student Answer: A
Correct Answer: B
Marks: 0/1

11. TIBCO Spotfire is best classified as which type of tool?
A) ETL tool
B) CI/CD pipeline orchestrator
C) Business Intelligence (BI) and Data Visualization platform
D) NoSQL database management system
Student Answer: C
Correct Answer: C
Marks: 1/1

12. Google BigQuery is designed around which core architectural principle to achieve high scalability and query performance?
A) Massively parallel processing (MPP) columnar storage.
B) Traditional row-based indexing.
C) Sharding data across independent transactional servers.
D) In-memory graph database model.
Student Answer: B
Correct Answer: A
Marks: 0/1

13. In data cleaning, what statistical method is commonly used to identify potential outliers based on their distance from the quartiles?
A) Z-score standardization
B) Min-Max scaling
C) Interquartile Range (IQR) method
D) Box-Cox transformation
Student Answer: C
Correct Answer: C
Marks: 1/1

14. Which type of modeling is most appropriate for pattern detection aimed at segmenting customers into distinct, unlabeled groups based on purchasing behavior?
A) Regression (Supervised)
B) Clustering (Unsupervised)
C) Classification (Supervised)
D) Reinforcement Learning
Student Answer: D
Correct Answer: B
Marks: 0/1

15. Which technique is specifically used during ML modeling training to prevent a model from becoming overly complex and performing poorly on unseen data?
A) Normalization
B) Dimensionality Reduction
C) Regularization (L1 or L2)
D) Feature Scaling
Student Answer: C
Correct Answer: C
Marks: 1/1

16. Which library is the standard Python choice for creating interactive, web-based visualizations without writing JavaScript?
A) Matplotlib
B) Seaborn
C) Plotly
D) Pillow
Student Answer: A
Correct Answer: C
Marks: 0/1

17. What is Flask primarily categorized as within the realm of web development?
A) A full-stack MVC framework
B) A micro-framework
C) A database ORM tool
D) A JavaScript front-end library
Student Answer: D
Correct Answer: B
Marks: 0/1

18. In Flask applications, which component is typically used to manage user sessions and securely store small amounts of data across requests?
A) The request object
B) The Jinja template engine
C) Cookies (via the 'session' object)
D) The application context
Student Answer: D
Correct Answer: C
Marks: 0/1

19. MongoDB uses which database model, making it a highly flexible NoSQL solution?

A) Relational (Table-based)
B) Key-Value
C) Document
D) Graph
Student Answer: A
Correct Answer: C
Marks: 0/1

20. Artificial Intelligence systems that are designed to perform a narrow task (e.g., Siri, AlphaGo) are classified as:
A) Artificial General Intelligence (AGI)
B) Artificial Super Intelligence (ASI)
C) Narrow AI (Weak AI)
D) Turing Complete AI
Student Answer: A
Correct Answer: C
Marks: 0/1

21. The machine learning algorithm known as K-Means falls under which category?
A) Supervised Learning
B) Reinforcement Learning
C) Unsupervised Learning
D) Semi-Supervised Learning
Student Answer: A
Correct Answer: C
Marks: 0/1

22. The Xception Convolutional Neural Network architecture is known for replacing standard convolutions with which operation to improve performance and efficiency?
A) Fully connected layers
B) Depthwise Separable Convolutions
C) Pooling layers
D) Recurrent layers
Student Answer: D
Correct Answer: B
Marks: 0/1

23. In Python, what is the key characteristic of a 'generator' compared to a 'list'?
A) Generators store all items in memory immediately.
B) Generators are mutable, while lists are immutable.
C) Generators calculate and yield items one at a time (lazy evaluation).
D) Lists must use the 'yield' keyword.
Student Answer: B
Correct Answer: C
Marks: 0/1

24. What is the primary purpose of a 'foreign key' constraint in a relational SQL database?
A) To guarantee that all column values are unique.
B) To speed up data retrieval using indexing.
C) To enforce referential integrity between two tables.
D) To define the primary identifier for a table.
Student Answer: D
Correct Answer: C
Marks: 0/1

25. In the C language, if `int *p;` and `int a = 10;`, what is the result of the expression `*p = a;` if `p` has not been initialized?
A) The value of 'a' (10) is stored in a valid memory location.
B) A compiler warning is issued, but the program runs normally.
C) The statement is illegal in C.

D) The program results in undefined behavior or a segmentation fault.
Student Answer: C
Correct Answer: D
Marks: 0/1

FINAL SCORE: 5/25